

Unit 1 Practice — The Real Number System

Classifying · Converting · Estimating Square & Cube Roots · Ordering Numbers

Score: 0 | Answered: 0 / 39

Part A — Classify Each Number

 **Remember:** Rational = can be written as a/b (fractions, terminating decimals, repeating decimals).
Irrational = non-repeating, non-terminating decimals.

Question 1

Classify this number:

$0.333\dots$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 2

Classify this number:

π

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 3

Classify this number:

$\sqrt{49}$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 4

Classify this number:

$\sqrt{10}$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 5

Classify this number:

$-3/8$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 6

Classify this number:

$0.1212\dots$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 7

Classify this number:

$\sqrt{144}$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 8

Classify this number:

$0.101001\dots$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 9

Classify this number:

$2/7$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 10

Classify this number:

$$\sqrt{2}$$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 11

Classify this number:

$$-\sqrt{81}$$

RATIONAL or IRRATIONAL?

Rational

Irrational

Question 12

Classify this number:

$$3.14$$

RATIONAL or IRRATIONAL?

Rational

Irrational

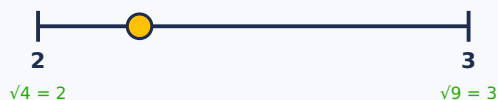
Part A — Estimate Each Square or Cube Root

 **Algorithm:** 1) Find the perfect squares (or perfect *cubes*) above & below. 2) Determine which is closer. 3) Estimate to the nearest tenth. A leading – just flips the sign.

Question 1

Estimate $\sqrt{5}$

$$\sqrt{5} \approx ?$$



2.2

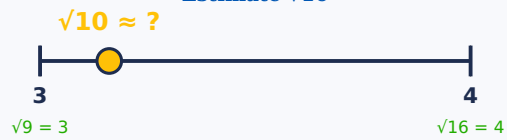
2.0

2.5

2.6

Question 2

Estimate $\sqrt{10}$



3.5

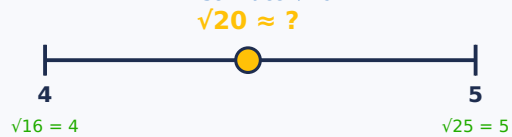
3.6

3.0

3.2

Question 3

Estimate $\sqrt{20}$



4.9

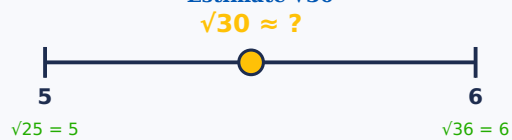
4.5

4.2

4.0

Question 4

Estimate $\sqrt{30}$



5.0

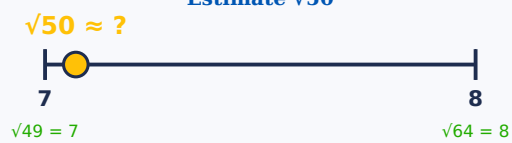
5.5

5.2

5.9

Question 5

Estimate $\sqrt{50}$



7.1

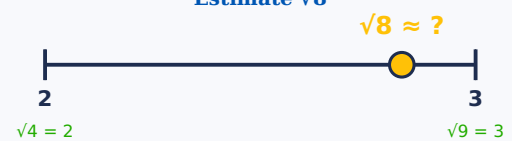
7.5

7.0

6.8

Question 6

Estimate $\sqrt{8}$



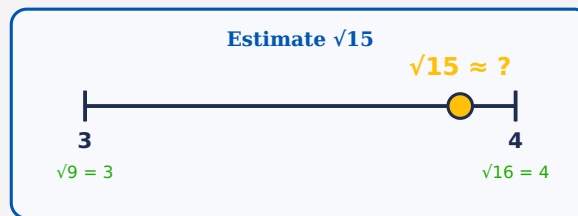
2.8

2.4

2.5

2.0

Question 7



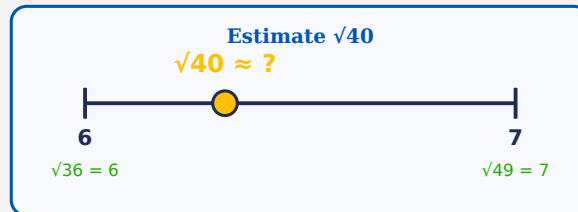
3.6

3.5

3.9

3.0

Question 8



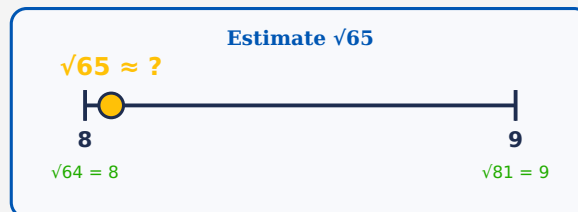
6.0

6.3

6.7

6.5

Question 9



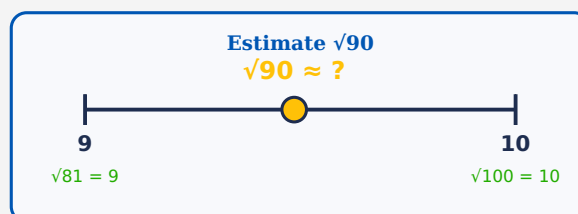
8.1

8.5

8.0

7.8

Question 10



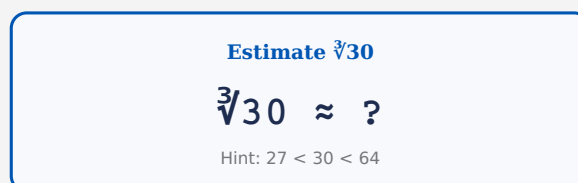
9.2

9.5

9.9

9.0

Question 11



3.1

3.9

5.5

2.7

Question 12

Estimate $\sqrt[3]{100}$

$$\sqrt[3]{100} \approx ?$$

Hint: $64 < 100 < 125$

4.6

5.0

10.0

3.3

Question 13

Estimate $-\sqrt{40}$

$$-\sqrt{40} \approx ?$$

Hint: estimate $\sqrt{40}$ first, then flip the sign

-6.3

6.3

-5.5

-20

Part A — Order from Least to Greatest

 **Strategy:** Convert each number to a decimal approximation, then arrange from least to greatest. Click numbers in order (smallest first), then check your answer!

Question 1

Order from LEAST to GREATEST: π $\sqrt{10}$ $10/3$ 3.5

π

$\sqrt{10}$

$10/3$

3.5

← Least

Greatest →

1st

?

2nd

?

3rd

?

4th

?

Reset

Check Answer

Question 2

Order from LEAST to GREATEST: 1.5 $\sqrt{2}$ $7/4$ $\sqrt{3}$

1.5

$\sqrt{2}$

$7/4$

$\sqrt{3}$

← Least

Greatest →

1st

?

2nd

?

3rd

?

4th

?

Reset

Check Answer

Question 3

Order from LEAST to GREATEST: $\sqrt{5}$ 2.5 $\frac{11}{5}$ $\sqrt{7}$

2.5	$\frac{11}{5}$	$\sqrt{7}$	$\sqrt{5}$
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← Least

Greatest →

1st	2nd	3rd	4th
?	?	?	?

↶ Reset

Check Answer

Question 4

Order from LEAST to GREATEST: 4.2 $\sqrt{18}$ $\frac{13}{3}$ $\sqrt{20}$

$\frac{13}{3}$	$\sqrt{20}$	4.2	$\sqrt{18}$
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← Least

Greatest →

1st	2nd	3rd	4th
?	?	?	?

↶ Reset

Check Answer

Question 5

Order from LEAST to GREATEST: -2.5 $-\sqrt{5}$ $-\frac{8}{3}$

-2.5	$-\frac{8}{3}$	$-\sqrt{5}$
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← Least

Greatest →

1st	2nd	3rd
?	?	?

↶ Reset

Check Answer

Question 6

Order from LEAST to GREATEST: $\sqrt{50}$ 7.1 $\frac{22}{3}$ $\sqrt{52}$

7.1	$\sqrt{52}$	$\frac{22}{3}$	$\sqrt{50}$
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← Least

Greatest →

1st	2nd	3rd	4th
?	?	?	?

↶ Reset

Check Answer

Question 7

Order from LEAST to GREATEST: $\frac{5}{6}$ $\sqrt{0.5}$ 0.9 $\frac{7}{8}$

$\frac{5}{6}$	$\sqrt{0.5}$	0.9	$\frac{7}{8}$
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← Least

Greatest →

1st	2nd	3rd	4th
?	?	?	?

Reset

Check Answer

Question 8

Order from LEAST to GREATEST: $\sqrt{30}$ 5.5 $17/3$ $\sqrt{32}$

$\sqrt{30}$	5.5	$\sqrt{32}$	$17/3$
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← Least

Greatest →

1st	2nd	3rd	4th
?	?	?	?

Reset

Check Answer

Part A — Convert Between Fractions & Decimals

Remember: Fraction → decimal = divide top by bottom. Repeating decimal → fraction = put the repeating block over that many 9's, then simplify (one digit → /9, two digits → /99).

Question 1

Write as a decimal:

$5/8$

0.625

0.85

0.58

0.65

Question 2

Write as a fraction (simplest form):

0.6

$3/5$

$6/10$

$2/3$

$1/6$

Question 3

Write as a fraction (simplest form):

0.4444...

(the 4 repeats)

$4/9$

$2/5$

$4/10$

$1/4$

Question 4

Write as a fraction (simplest form):

0.7272...

(the digits 72 repeat)

8/11

7/9

72/100

18/25

Question 5

Write as a decimal:

7/20

0.35

0.72

0.720

0.035

Question 6

Write as a fraction (simplest form):

0.125

1/8

1/4

1/5

5/8

Study Tools

Worked Examples

Click "Show Next Step" to reveal solutions one step at a time.

Estimate $\sqrt{45}$

Step 1: Find perfect squares around 45

$$36 < 45 < 49 \rightarrow \sqrt{36} < \sqrt{45} < \sqrt{49}$$

Show Next Step →

Convert 0.27 to a fraction

Step 1: Count repeating digits

27 repeats → 2 digits

Show Next Step →

🔍 Common Mistakes

Can you spot what went wrong?

Classify: 0.333...

✗ **Student:** Student says: Irrational (it goes on forever)

✓ **Fix:** WRONG! Repeating decimals ARE rational. $0.333... = 1/3$

💡 Irrational means non-repeating AND non-terminating.

Estimate $\sqrt{20}$

✗ **Student:** Student says: $\sqrt{20} \approx 10$ (because $20/2 = 10$)

✓ **Fix:** WRONG! You don't divide by 2. $\sqrt{20}$ is between $\sqrt{16}=4$ and $\sqrt{25}=5$, so ≈ 4.5

💡 Find the perfect squares AROUND the number, not divide.

🔑 Key Tips for This Unit

- ✓ All fractions are rational — even ugly ones like $17/31$.
- ✓ π is NOT 3.14 — 3.14 is rational. π itself is irrational.
- ✓ If the square root is not a perfect square, it's irrational.
- ✓ Memorize perfect squares 1-144: 1,4,9,16,25,36,49,64,81,100,121,144
- ✓ Cube roots: $\sqrt[3]{}$ asks for THREE equal factors — $\sqrt[3]{64} = 4$ (not 8). Memorize cubes: 1,8,27,64,125,216,343,512,729,1000